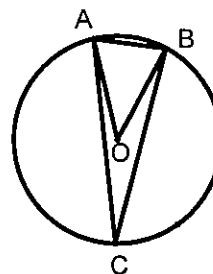


9. In vector notation, $\vec{v} = \langle a, b \rangle = a\vec{i} + b\vec{j}$. Determine the degree measure of the angle between the vectors $\vec{u} = \langle 2, 3 \rangle$ and $\vec{v} = \langle -2, 5 \rangle$. Express your answer as a decimal rounded to the nearest hundredth.

10. The probability a certain man will be alive 25 years hence is $\frac{3}{7}$ while independently, the probability his wife will be alive 25 years hence is $\frac{4}{5}$. Determine the probability that at least one of this man and woman is alive 25 years hence. Express your answer a common fraction reduced to lowest terms.

11. In the figure shown, the radius of Circle O is 8 and $AB = 6$. Determine the degree measure of $\angle C$. Express your answer as a decimal rounded to the nearest hundredth.



12. If $x > 5$, determine the largest value of k so that $\log(x) - k = \log(x-5) + 3$ has no real solutions.

13. $\frac{4x^4 + 2x^3 + 5x^2 + 15}{x+5} = Q(x) + \frac{k}{x+5}$. Determine the value of k .

14. $\angle A$ and $\angle B$ are first quadrant angles such that $\sin A = \frac{4}{7}$ and $\cos B = \frac{5}{7}$. In reduced and simplified radical form, $\tan(A+B) = \frac{k\sqrt{p} + w\sqrt{q}}{f}$ where k , w , p , q , and f are integers. Determine the value of the sum $(k + w + p + q + f)$.